

User Guide MetaboAnalyst 6.0

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1. Introduction

This document guides researchers on using MetaboAnalyst 6.0 [1] for functional, enrichment, pathway, and network analysis in metabolomics. It outlines how MetaboAnalyst can be used effectively by providing step-by-step instructions and helping users to utilize the platform with little/without prior knowledge. Only options that can be used with non-targeted output from Compound Discoverer (CD) or the shiny app are discussed. All other options require input from targeted analysis or other omics platforms.

Specific questions that might come up throughout the analysis are listed in section 8 (FAQ).

2. Overview of MetaboAnalyst

Open [MetaboAnalyst 6.0](#). You can find specific tutorials [here](#).

2.1 What is MetaboAnalyst

MetaboAnalyst 6.0 is a web-based platform that provides several tools for comprehensive metabolomic data analysis. These analyses help identify biologically relevant pathways/metabolite sets crucial for understanding metabolic processes and their implications in various research fields.

The Module Overview on the website shows all available steps for data processing and useful utilities that help streamlining the input.

Input Data Type: **MS Peaks** allows you to apply **Functional Analysis** with data directly exported from CD on a feature basis.

Input Data Type: **Annotated Features** allows you to input MetaboAnalyst IDs from the Shiny app in three different ways:

- Enrichment Analysis
- Pathway Analysis
- Network Analysis

2.2 Difference between Functional, Enrichment, Pathway, and Network Analysis

Functional analysis focuses on the biological roles or processes represented in a dataset. Enrichment analysis identifies biological pathways that are statistically over-represented in a dataset. Pathway analysis maps metabolites to known biological pathways, providing insights into metabolic functions and interactions. Network analysis helps visualize the complex relationships among metabolites and can provide insights into regulatory mechanisms.

In short:

- Functional → What is there?
- Enrichment → What occurs more than expected?
- Pathway → Where do changes happen?
- Network → How are things connected?

In terms of **data input**, functional analysis in MetaboAnalyst 6.0 works with true non-targeted data. The other three options are mainly used with targeted data, but each category has one input option that also works for level 1 or level 2 annotation lists of non-targeted data from CD ideally with background metabolites.

Either the [mummichog algorithm](#) [2] or GSEA (gene set enrichment analysis) [3] can be selected in functional analysis, while MSEA (metabolite set enrichment analysis) [4] is used in enrichment analysis.

3. Functional Analysis

Functional analysis is particularly designed for non-targeted data. You can generate the specific files directly from CD. Different input options are feasible. Check FAQ number 5.

Three different options for uploading data are available:

- A peak list profile
- A peak intensity table
- MS Peak and Annotation list

Try the different example datasets to see what the input and output is.

3.1 A peak list profile

3.1.1 Please upload your data

- Ion mode: mixed
 - Can also be positive or negative
- Mass Tolerance (ppm): 3
 - Depends on your instrument
- Retention time: not present
 - Can be used in version 2 (V2) to align different adducts
- Ranked by (1 column only): p-value
- Enforce Primary Ions (V2 only): checked
- Data File: +Choose file
 - The file is either a 1 column file where m/z are ranked by p-value or it has 3 or 4 columns. m.z, p.value, and t.score or fold change are mandatory. Retention time (rt) and mode are optional, but mode is recommended.

3.1.2 Data integrity check

Check all parameters that were selected and if the data you uploaded is correct.

3.1.3 Parameter Settings

Before proceeding, the analysis parameters, the chosen pathway library, and the minimum number of entries required per pathway or metabolite set must be defined.

Specify analysis parameters

- Algorithms
 - Mummichog
 - P-value cutoff: 0.05 or the filter you want to use based on your data
 - Which version: select 2.0 especially when using retention times
 - GSEA
 - Use this for overall rank based on t.score (fold change)
 - Cannot be selected for one column input
- Visual analytics
 - Scatter plot
 - Heatmaps
 - Only for multiple groups in the peak intensity table input
- Advanced options
 - Edit currency metabolites
 - Edit adducts

Select a pathway library

- Mammals
- Birds
- Fish etc.
- Metabolite Sets

3.1.4 Results

Depending on the parameter selection the result is either a Mummichog Pathway Activity Profile or a GSEA Pathway activity profile.

3.1.4.1 Mummichog Pathway Activity Profile

The $-\log_{10}(\text{P-value})$ is plotted against the enrichment factor of significant pathways. In addition, a detailed result table is provided.

The plot, pathway hits, and compound hits can be downloaded in different formats. In addition, the Enrich Network and the KEGG Network can be further explored. Different settings can be changed when exploring the networks and the graphic and result table can be downloaded.

3.1.4.2 GSEA Pathway Activity Profile

The $-\log_{10}(\text{P-value})$ is plotted against the Normalized Enrichment Score (NES). Horizontal displacement reflects enrichment strength and direction, while vertical position indicates statistical significance.

The NES answers how strong the enrichment is relative to what I would expect by chance for a metabolite set of this size.

- NES > 0: gene set enriched in the upregulated direction
- NES < 0: enriched in the downregulated direction

In addition, a detailed result table is provided as well.

3.1.5 The same downloads and networks as in the Results

Depending on the parameter selection the result is either a Mummichog Pathway Activity Profile or a GSEA Pathway activity profile.

Mummichog Pathway Activity Profile in 3.1.4.1 can be selected.

3.2 A peak intensity table

This tab allows you to upload a table that holds an intensity for each feature and sample. It does not support both polarities loaded at once, but only positive or negative ion mode. The data source is generic when files from “Thermo Scientific” machines are used.

A data integrity check is necessary to guarantee the correct format.

3.2.1 Data Filtering

It is recommended to filter the data already in CD to have a better control of your data set and omit these thresholds.

3.2.2 Missing value estimation

Upon filtering your data in CD, there should be no missing values.

3.2.3 Normalization overview

Select None for all 3 normalization parameters as data is already normalized by CD.

3.2.4 Parameter Settings

See 3.1.3.

GSEA cannot be selected when there are more than two groups

Now, also the heatmap can be selected as multiple groups are usually submitted. Clustering of the samples can also be seen in CD.

3.2.5 Results

See section 3.1.4.1.

When “heatmap” is selected in the parameters section a heatmap is shown instead of the “Mummichog Pathway Activity Profile”. It can be modified with respect to resolution, colors, borders, and how to cluster the peaks. The plot and results can be downloaded.

3.3 MS Peak and Annotation lists

This tab requires a peak table and an annotation file.

3.3.1 Upload peak and compounds lists

- Ion mode: Mixed mode
 - Optional: negative mode, positive mode
- Mass Tolerance (ppm): 3 ppm
- Annotation ID type: InchiKeys
 - Optional: HMDB ID, PubChem CID, PubChem SID, SMILES
- Enforce Primary Ions (V2 only): check
- Peak Data File and Annotation File:
 - See example datasets for format

3.3.2 Data integrity Check

See 3.1.2. Click Proceed

3.3.3 Parameter Setting

See 3.1.3.

3.3.4 Results

See 3.1.4.

4. Enrichment Analysis

Metabolite set enrichment analysis (MSEA) [4] is performed in enrichment analysis. MSEA is based on GSEA that was developed about two decades ago. [3]

Enrichment Analysis has 3 options with different input:

- **Over Representation Analysis**
 - List of compounds
- **Single Sample Profiling**
 - List of compounds with concentrations
- **Quantitative Enrichment Analysis**
 - Concentration table

The MetaboAnalyst output of the shiny app can only be used for option 1 – Over Representation Analysis.

4.1 Over Representation Analysis (ORA)

This option works with a list of compound IDs.

4.1.1 Input data

The MetaboAnalyst output from the shiny app is a single column text file for each comparison that contains **HMDB and KEGG IDs of significant features**. Each list can be used as is.

- Please enter a one-column compound list:
 - Copy/paste the list from the shiny app
 - Or paste any other ID list of significant features
- Input Type: HMDB and KEGG ID
 - Optional: Compounds name, HMDB ID, KEGG ID, PubChem CID, ChEBI ID, METLIN
- Feature Type: General
 - Optional: metabolites, lipids
- Try Example: None
 - Optional: List 1-3 of example data
- Submit

4.1.2 Name/ID Standardization

The compound selection is well described on the website. There will always be some IDs that are not in the database. Check these compounds manually and replace them with appropriate IDs if possible. Try to consider stereochemistry.

- Check compound IDs
 - Correct IDs that were not correctly matched, if possible
- Results can be downloaded
 - Link is at the bottom left side
- Proceed

4.1.3 Parameter setting

The enrichment tests and metabolite sets are described on the website and the FAQ in section 8. Note the difference of metabolomics compared to transcriptomics.

Different decisions need to be made here:

- Select metabolite set libraries
- Decide how many entries a metabolite set needs to contain
- Specify if a reference metabolome should be used
- Submit

Choose one metabolite set from the listed groups:

- Pathway base
- Disease signatures
- Chemical structures
- Other types
- Self defined

It is important to **specify whether a reference metabolome is used or not**. This influences the results. Two options are possible to select:

1. Use all the compounds in the selected library
 - Upload only significantly changed metabolites
 - MetaboAnalyst assumes a generic background (i.e. all metabolites in the database/pathways)
 - This is simple, but not ideal for metabolomics, because non-targeted LC-MS does not measure all metabolites equally

2. Upload a reference metabolome based on your analytical platform
 - the reference file contains all metabolites that could be detected by your platform, regardless of whether they changed or not
 - this defines the correct background for ORA
 - enrichment results are then driven mainly by biological regulation, not platform or coverage bias

When you want to upload a reference metabolome (option 2) with data generated by the shiny app you first need set the statistical parameter to 0 and change the project name (e.g. background) to additionally export all IDs without statistical restrictions. Then, the KEGG and HMDB IDs need to be converted to compound names. Use the [Compound ID Conversion tool](#) because it guarantees that the spelling aligns with MetaboAnalyst.

4.1.4 Results of Enrichment Analysis

The results section has different output option:

- Bar Chart
- Network View
- Dot plot
- Metabolite Set

At the bottom of the page, the results can be submitted for downloading the data.

4.1.4.1 Bar Chart

The bar chart shows the enrichment ratio plotted for the top 25 metabolite sets. Hovering over a bar shows you the exact enrichment ratio per set. They are ranked from top to bottom from most significant to least significant.

- bar length = how strong the pathway-level signal is relative to the background
- color = how statistically significant that enrichment is

4.1.4.2 Network View

All metabolite sets (nodes) that are enriched are shown in this network plot. Not all nodes need to be connected. Some sets have only one connection while some have multiple ones. The nodes can be dragged to change the position and a double click on a node shows you the metabolite set name with and highlights matched metabolites from your input in red. In addition, the reference is listed, can be selected, and leads to the specific metabolite set.

- dot size = enrichment ratio
- color = how statistically significant that enrichment is

4.1.4.3 Dot Plot

The dot plot gives an overview of the top 25 enriched metabolite sets. They are listed from most significant to least significant from top to bottom. The x-axis shows the $-\log_{10}$ p-values. The ratio of the dot represents the enrichment ratio. The larger it is, the higher the enrichment ratio is.

4.1.4.4 Metabolite Set list

This list shows all metabolite sets that were enriched. It lists

- total number of metabolites
 - how many metabolites are in the set
- number of hits
 - how many metabolites of my data (input) can be found in each set
- expect
 - The expected number of hits by chance, given:
 - the size of your input list
 - the size of the pathway
 - the background metabolite universe
 - If your input list is small compared to the total metabolite universe, that average is usually < 1 .
- p-value
 - raw enrichment p-value, from either:
 - hypergeometric test
 - Fisher's exact test
- Holm P
 - Holm–Bonferroni corrected p-value, which:
 - controls family-wise error rate (FWER)
 - is more conservative than FDR
 - Use this if you want very strict multiple-testing correction.
- FDR
 - False Discovery Rate–adjusted p-value (Benjamini–Hochberg)
 - Among significant pathways, how many are expected to be false positives?
 - In metabolomics, FDR is usually preferred over Holm
- View
 - Link to the metabolite set

4.1.5 Download Results & Start New Journey

At last, all plots and lists can be downloaded. In addition, a report can be generated but this function is no longer maintained and is outdated. The settings of the plots can be changed in the results section 4.1.4.

5. Pathway Analysis

132 different libraries can be visualized integrating enrichment analysis and pathway topology analysis.

Pathway analysis has 5 options with different input:

- **Compound List**
 - List of Compounds
- **Concentration Table**
 - Concentration data in .csv or .txt
- **Metabolomics Workbench Data**
 - Study ID from the Metabolomics Workbench
- **Metabolon Data sheet**
 - Xlsx file from Metabolon
- **Joint Pathway Analysis**
 - Significant genes or proteins with optional FC
 - Txt file with complete peak list with m/z and names

The output of the shiny app can only be used for option 1 – Compound List.

5.1 Compound List

This option works with a list of compound IDs.

5.1.1 Input data

The output from the metabolomics shiny app is a single column text file that contains **HMDB and KEGG IDs of significant features**.

- Enter a one-column compound list:
 - Copy/paste the list from the shiny app
 - Or input any other list of significant features
- Input Type: HMDB ID
 - Optional selections: Compound name, KEGG ID
- Submit

5.1.2 Name/ID Standardization

Same as 4.1.2

When entering a mixed list of input data (HMDB ID and KEGG ID) please view each KEGG ID and match the corresponding HMDB if possible.

5.1.3 Parameter setting

Similar to enrichment analysis, a reference metabolome is recommended to perform pathway analysis. Upon selecting the method and library of choice press the submit button.

Multiple combinations are possible when **specifying** the pathway **analysis parameters**:

- **Visualization** method
 - Scatter plot
 - Great for spotting pathways that are both significant and impactful
 - Heatmap
 - Cannot be selected so far
- **Enrichment** method
 - Hypergeometric Test
 - Classic over-representation test
 - Assumes random sampling without replacement
 - Fisher's Exact Test
 - Similar to the hypergeometric test
 - For small datasets
- **Topology** measure
 - Relative-betweenness Centrality
 - How often lies a metabolite on the shortest paths between other metabolites
 - Sensitive to pathway structure
 - Out-degree Centrality
 - How many downstream connections does a metabolite have
 - Less sensitive to global pathway structure than betweenness
- **Reference** metabolome
 - Check 4.1.3

In addition, a pathway **library** that is suitable for the data needs to be selected from the following groups:

- Mammals
- Birds
- Fish etc.

5.1.4 Result view

How to use the result view is well described on the website.

When the scatter plot is selected, it gives an overview of the pathways from enrichment analysis where the $-\log_{10}$ p-value is plotted against the pathway impact. Each pathway can be viewed separately with significant features highlighted in red.

The “Enrichment network” button redirects to a separate site to have a deeper look at the network. At the bottom of the site is a list of all perturbed pathways.

The results can be downloaded upon hitting the submit button.

5.1.5 Download Results & Start New Journey

See 4.1.5.

6. Network Analysis

Network analysis helps to visualize the complex relationships among metabolites and can provide insights into regulatory mechanisms.

The molecules of interest can be visually explored with respect to biological networks such as the KEGG global metabolic network. In addition, networks based on known associations between genes, metabolites, and diseases can also be investigated.

Network analysis has 2 options with different input:

- **List of genes/compounds**
 - List of genes (human only) or KEGG orthologs
 - Log2Fold changes are optional
 - List of metabolites
 - Log2Fold changes are optional
- A concentration table
 - File with concentrations
 - Or Study ID from Metabolomics Workbench

The output of the shiny app can only be used for option 1 – List of genes/compounds.

6.1 List of genes/compounds

6.1.1 Input data

This works with a **list of genes** (human only) or KEGG orthologs **in combination with metabolites**. Log2 fold changes can be entered optional. Metabolite input can be compound names, HMDB ID, or KEGG ID. Combined input is not possible so far, but KEGG IDs can be converted into HMDB IDs when not found as in 5.1.2.

Consider the **different ID Type inputs**. Many different combinations are possible.

Try the example to see what the input looks like.

6.1.2 ID Mapping

Similar to Name/ID standardization (4.1.2). Check if all input data is correctly mapped. There is a tab for

- Compound Name Mapping
- Gene Name Mapping

6.1.3 Network Analysis Options

5 different network options can be selected. They are all well described on the website.

- KEGG Global Metabolic Network
- Metabolite-Disease Interaction Network
- Gene-Metabolite Interaction Network
- Metabolite-Metabolite Interaction Network
- Metabolite-Gene-Disease Interaction Network
- Debiased Sparse Partial Correlation (DSPC) Network
 - Cannot be selected so far

7. Conclusion

MetaboAnalyst 6.0 is an essential tool for gaining insights from metabolomic data. Through this guide, users get an overview of functional, enrichment, pathway, and network analyses to enhance their research outcomes.

8. FAQ

1. Which questions would you like to answer:

- **Functional Analysis**
 - Which biological functions, molecular processes, metabolite classes (e.g. amino acids, organic acids, lipids) are affected in my dataset?
- **Enrichment Analysis**
 - Are certain pathways or functional categories statistically over-represented compared to what would be expected by chance?
 - Do significantly changed metabolites cluster in specific biological processes?
 - Are altered metabolites enriched in specific metabolic routes or compound classes?
- **Pathway Analysis**
 - Which metabolic pathways are perturbed and in what direction (up/down)?
 - Where in the pathway do changes occur, and how large are their effects?
 - Which pathways show coordinated perturbation rather than isolated metabolite changes?
 - Which metabolic pathways are disrupted under the experimental condition?
 - Are changes localized to specific pathway branches or rate-limiting steps?
- **Network Analysis**
 - How are metabolites functionally or statistically connected to each other?
 - Are there key hub metabolites or subnetworks associated with the phenotype or outcome?
 - Which nodes have high centrality or control information flow in the network?
 - Which metabolites show strong co-variation across samples?

2. What is a feature:

A feature is a signal detected by the data processing software and does not yet represent a uniquely identified molecule. It has a specific m/z (mass to charge) value with a specific retention time and an intensity. The m/z usually refers to a (single) charged ion (e.g. $[M+H]^{+1}$ or $[M-H]^{-1}$) detected by the software. In addition to (de-)protonation, other adducts (e.g. +Na, +K, +NH₄) or losses (e.g. -H₂O) are also possible, each resulting in a separate feature.

3. What is a metabolite/compound:

A compound is a molecule that, when measured by MS, can give rise to one or more features characterized by specific m/z values, retention times, and isotopic patterns. From

the exact mass and the isotopic pattern, a chemical formula can be calculated. A metabolite is a small molecule (< 1500 Da) and part of metabolism (intermediate or end product). Due to the chemical properties of a metabolite/compound, it often is detected in the MS as multiple features (adducts and/or losses). The software (e.g. CD) assembles these features into one exact mass (or m/z) with a specific isotopic pattern if possible. Sometimes, features that belong to the same compound cannot be merged. This is especially true, if (in-source) fragmentation occurs in the MS. **Note: exact mass $\neq$ m/z**

4. What means annotated:

Annotation assigns a putative identity or name to a measured compound. Depending on the available evidence (exact mass, isotopic pattern, MS/MS, and retention time), different confidence levels can be reached (MSI levels 1–4). This name is based on the exact mass, the chemical formula, and ideally on the MS/MS (fragmentation pattern – level 2) and additionally the retention time (level 1). Once a compound is annotated, database identifiers (e.g. KEGG, HMDB) can be assigned. **Note: annotation $\neq$ identification**

According to the Metabolomics Standards Initiative (MSI) [5]:

- **Level 1** = confirmed identification (standard, RT, MS/MS)
- **Level 2** = putative annotation (MS/MS match)
- **Level 3** = putative compound class (exact mass found in database)
- **Level 4** = unknown (exact mass)

5. What is the difference between the 3 Functional Analysis options:

The difference lies in the **data input** and which options that you can further select

- **A peak list profile**
 - Data Input: Unannotated features, p-values, t.scores/fold changes, optional rt and mode or one m/z column ranked by p-value
 - Functional analysis: Mummichog or GESA
 - Visualization: Scatterplot
 - Key point: Works with raw feature lists before annotation.
- **A peak intensity table**
 - Data Input: Unannotated features with intensities across samples (for group comparison), optional QC and blank samples
 - Functional analysis: Mummichog only
 - Visualization: Scatterplot or Heatmap

- Key points:
 - GSEA is not available here because GSEA requires ranked statistics across groups (like log2FC or p-values).
 - You can generate heatmaps because you have sample-level intensities.
- **MS Peak and Annotation list**
 - Data input: Annotated features, p-values, t.scores/fold changes, optional rt and mode
 - Functional analysis: Mummichog or GSEA
 - Visualization: Scatterplot

6. Should I use Mummichog or GSEA for Functional Analysis:

Use **Mummichog** if:

- You have untargeted metabolomics data
- Most features are m/z values, not confidently identified metabolites
- You expect many false or ambiguous annotations
- You're working at feature level, not compound level

Why: It:

- works directly on m/z features
- tolerates wrong/missing annotations
- looks for pathway-level structure that rises above noise

If you don't have features that match into solid IDs, GSEA might not be optimal.

Use **GSEA** if:

- Features are matched to putative metabolites
- You trust the ranking metric (log2FC, t-statistic, etc.)
- You want directionality (up vs. down enrichment)
- Your metabolite list is relatively clean

Why: GSEA assumes the identities/m/z are real. Garbage IDs → garbage enrichment.

Running both is very common but:

- Expect partial overlap, not identical pathways
- Trust directionality only from GSEA
- Trust discovery more from Mummichog

- Highlight consensus pathways as strongest evidence

7. What is the difference between the scatter plot and the heatmap

- Heatmap = What are the features doing across samples?
- Scatter plot = Which pathways emerge as both strong and reliable given those feature patterns?

Heatmaps:

- Visual pattern recognition
 - You can see if groups of features are consistently higher/lower in certain samples
 - Helps detect batch effects, outliers, or strongly regulated feature clusters
- Linking to pathways
 - While individual rows aren't annotated, clusters of features that show the same pattern may drive pathway enrichment seen in mummichog plots
 - In other words, it gives you context for why a pathway is enriched: which features are contributing most
- Data exploration/QC
 - Before trusting pathway results, it's useful to see the raw feature behavior across samples

Scatter plot:

- Effect size vs. significance overview
 - Displays pathways in terms of magnitude of change versus statistical confidence
 - Makes it easy to compare pathways against each other in a single view
 - Quickly identify pathways that are both strongly affected and statistically robust
 - Pathways in the top-right region (high activity + high significance) are most likely to be biologically meaningful
 - Helps distinguish pathways with many weakly changing features from those driven by coherent, strong signals
 - Reveals trade-offs (e.g. highly significant but low-impact pathways vs. high-impact but borderline-significant ones)
- Relationship between feature-level signal and pathway enrichment
 - Each point summarizes the aggregate contribution of many features mapped to a pathway

- Complement the heatmap by showing how strongly and how confidently those feature patterns translate into pathway-level effects

8. How does the enrichment test work:

Enrichment tests use the well-established global test method [6, 7] to assess whether a group of metabolites is associated with the outcome. For each metabolite set, the method calculates a single score. This approach has been shown to perform as well as, or better than, many other commonly used enrichment methods.

9. What are the metabolite sets:

Unlike transcriptomics, targeted metabolomics typically measures only a small fraction of the metabolome, and this fraction depends on the analytical platform and protocol used. As a result, not all metabolites in a pathway have the same chance of being detected, and enrichment results may reflect both technical (platform-specific) and biological effects. To focus on true biological changes, we strongly recommend uploading a reference metabolome that includes all metabolites measurable on your platform which come from the non-targeted analysis including also annotated features that are not significantly regulated.

10. What is the enrichment ratio:

In global test, the “enrichment ratio” is not a true over-representation ratio. In MetaboAnalyst it is used as a descriptive effect-size-like measure, roughly reflecting:

- how strong the pathway-level signal is relative to the background
- often driven by the average contribution of metabolites in the set
- Larger bar $\approx$ stronger overall association
- But it is not based on counts of significant metabolites
- Treat it as a ranking/magnitude indicator, not a probability or fold-change.

11. What does the network show in ORA:

This is a knowledge-based overlap network, not a causal or dynamic one.

- Nodes = pathways
 - Each circle is a metabolic pathway (e.g. Glycolysis, TCA cycle, Amino sugar metabolism).
- Edges (connections) = shared metabolites
 - Two pathways are connected if they share one or more metabolites.

A line (edge) between nodes mean that these two pathways overlap biochemically because they use the same compounds.

- **Node size** → enrichment ratio
- **Node color** → statistical significance (p-value)
 - Red = more significant
 - Yellow = less significant

So big red nodes = **important, significantly enriched pathways**.

12. What do the parameters in pathway analysis tell me:

- Visualization: How do you want to look at it
 - Which pathways pop out
 - Which ones are both significant and biologically impactful
- Enrichment: Did this pathway show up
 - Do I see more significant metabolites than I'd expect by chance
- Topology: Did important parts of it change
 - Did the "key" nodes in the pathway change, or just peripheral ones

13. Which parameters are usually chosen for pathway analysis:

- Visualization: Scatter plot
- Enrichment: Hypergeometric Test
- Topology: Relative-betweenness Centrality

9. References

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